2.5 PreQuiz Solutions: Motion, Lab Skills, and Spreadsheets

1. Round each value to three significant figures.
   (a) 0.004817 m $\approx$ **0.00482 m**
   (b) 63.249 kg $\approx$ **63.2 kg**
   (c) 7.0036 s $\approx$ **7.00 s**
   (d) 21,560 N $\approx$ **21,600 N**

2. Perform each operation and give the answer in scientific notation, rounded to 3 significant figures.
   (a) $(3.2 \times 10^4) + (7.5 \times 10^3) = 32{,}000 + 7{,}500 = 39{,}500 = \mathbf{3.95 \times 10^4}$
   (b) $(8.40 \times 10^{-3}) \times (2.0 \times 10^4) = 8.40 \times 2.0 \times 10^{-3+4} = 16.8 \times 10^1 = \mathbf{1.68 \times 10^2}$
   (c) $\dfrac{6.3 \times 10^5}{9.0 \times 10^2} = \dfrac{6.3}{9.0} \times 10^{5-2} = 0.70 \times 10^3 = \mathbf{7.00 \times 10^2}$

3. Convert each value. Show one line of work using unit factors.
   (a) $36.0 \text{ km/h} \times \dfrac{1000 \text{ m}}{1 \text{ km}} \times \dfrac{1 \text{ h}}{3600 \text{ s}} = \mathbf{10.0 \text{ m/s}}$
   (b) $15.0 \text{ m/s} \times \dfrac{1 \text{ km}}{1000 \text{ m}} \times \dfrac{3600 \text{ s}}{1 \text{ h}} = \mathbf{54.0 \text{ km/h}}$

4. A student jogs along a straight track (forward is +). Start $x_0 = 5.0$ m.
   (a) $\Delta x = 17.0 - 5.0 = +\mathbf{12.0 \text{ m}}$
   (b) $\Delta x = 9.0 - 17.0 = \mathbf{-8.0 \text{ m}}$
   (c) Total displacement $= 9.0 - 5.0 = +\mathbf{4.0 \text{ m}}$
   (d) Total distance $= |+12.0| + |-8.0| = 12.0 + 8.0 = \mathbf{20.0 \text{ m}}$

Kinematics Equations

Formulas: $v = v_0 + at$ $\quad d = v_0 t + \frac{1}{2}at^2$ $\quad v^2 = v_0^2 + 2ad$ $\quad d = \frac{1}{2}(v_0 + v)t$

5. A bicycle accelerates uniformly from rest at 1.5 m/s^2.

   (a) $v = v_0 + at = 0 + (1.5)(4.0) = \mathbf{6.0\ m/s}$

   (b) $d = v_0 t + \frac{1}{2}at^2 = 0 + \frac{1}{2}(1.5)(4.0)^2 = \frac{1}{2}(1.5)(16) = \mathbf{12\ m}$

6. A ball is thrown vertically upward with an initial velocity of 12 m/s. Use $g = 10$ m/s^2.

   (a) At max height, $v = 0$. $\quad 0 = 12 - 10t \quad \Rightarrow \quad t = \mathbf{1.2\ s}$

   (b) $d = v_0 t + \frac{1}{2}at^2 = (12)(1.2) + \frac{1}{2}(-10)(1.2)^2 = 14.4 - 7.2 = \mathbf{7.2\ m}$

Lab and Technology Skills — Solutions

7. Graphical Analysis curve fit.

   The data follows $x = At^2 + Bt + C$. A quadratic fit gives approximately:

$$A \approx 0.160, \quad B \approx -0.040, \quad C \approx 0.20$$

   (a–b) Students create graph and report fit parameters from Graphical Analysis.

   (c) Since $A = \frac{1}{2}a$, we get $a = 2A = 2(0.160) = \mathbf{0.32\ m/s^2}$

   (d) File saved to Google Drive with student name.

8. Spreadsheet calculation.

   (a–d) Students paste screenshot, enter values, and write formula.

   (e) $F = ma = (0.390)(0.32) \approx \mathbf{0.12\ N}$

   Note: exact answer depends on student's curve fit parameters.